



Uropathogenic bacteria and deductive genomics towards antimicrobial resistance, virulence, and potential drug targets

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Abstract

Urinary tract infections (UTIs) are among the most prevalent bacterial infections affecting people in inpatient and outpatient settings. The current study aimed to sequence the genome of uropathogenic *Escherichia coli* strain CUI-B1 resourced from a woman having uncomplicated cystitis and pyelonephritis. Followed by deductive genomics towards potential drug targets using *E. coli* strain CUI-B1, strain O25b: H4-ST131, *Proteus mirabilis* strain HI4320, *Klebsiella pneumoniae* strain 1721, and *Staphylococcus saprophyticus* strain ATCC 15305 uropathogenic strains. Comparative genome analysis revealed that genes related to the survival of *E. coli*, *P. mirabilis*, *K. pneumoniae*, and *S. saprophyticus*, such as genes of metal-requiring proteins, defense-associated genes, and genes associated with general physiology, were found to be highly conserved in the genomes including strain CUI-B1. However, the genes responsible for virulence and drug resistance, mainly those that are involved in bacterial secretion, fimbriae, adherence, and colonization, were found in various genomic regions and varied from one species to another or within the same species. Based on the genome sequence, virulence, and antimicrobial-resistant gene dataset, the subtractive proteomics approach revealed 22 proteins mapped to the pathogen's unique pathways and among them, *entB*, *clbH*, *chuV*, and *ybtS* were supposed to be potential drug targets and the single drug could be utilized for all above-mentioned strains. These results may provide the foundation for the optimal target for future discovery of drugs for *E. coli*-, *P. mirabilis*-, *K. pneumoniae*-, and *S. saprophyticus*-based infections and could be investigated further to employ in personalized drug development.

Keywords Whole-genome sequencing · Virulence genes · Antibiotic resistance genes · Potential drug targets

Introduction

Urinary tract infections (UTIs) are among the most prevalent bacterial infections, affecting 150 million people annually around the globe, which mainly affects the bladder, urethra, ureters, and kidneys. UTIs can also be recurrent and lead to serious complications if left untreated (Klein and Hultgren 2020). UTIs are generally more frequent in women rather than in men, and risk hazards include sexual activity, pregnancy, menopause, urinary tract abnormalities, catheterization, and certain immune-compromising medical conditions (Shaeriya et al. 2021). The improper use of antibiotics has significantly contributed to the rise of multi-drug-resistant (MDR) bacteria, making it almost difficult to treat UTIs and other bacterial infections (Hatt and Rather 2008).

UTIs can be caused by various types of bacteria, but some of the most studied ones include *E. coli* accounting for about 80–85% of cases, *E. faecalis* mostly in people with

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underlying health conditions, *P. mirabilis* particularly in people with urinary tract abnormalities, *K. pneumoniae* particularly in people with weakened immune systems, and *S. saprophyticus* is a common cause of UTIs in young women. Other less common bacteria that can cause UTIs include *P. aeruginosa*, *Enterobacter* spp., and *S. marcescens* (Flores-Mireles et al. 2015; Hatt and Rather 2008). In the case of UTIs, bacteria can form biofilms on the walls of the bladder or urinary catheters, which can lead to recurrent or persistent infections. Biofilms can also provide a protective environment for bacteria, allowing them to evade the body's immune response and antibiotics. As a result, treating biofilm-associated UTIs can be challenging, and multiple rounds of antibiotics may be necessary to fully eliminate the infection (Stickler 2008; Pelling et al. 2019; Kahlmeter 2003).

Uropathogenic strains are distinguished by the expression of unique bacterial structures, properties, and products that are referred to as virulence factors, which facilitate the bacteria's invasion, colonization, and evasion of host defenses (Flores-Mireles et al. 2015; Stamm and Norrby 2001). These virulence factors include adhesins, such as type 1 fimbriae and P fimbriae, which allow bacteria to adhere and invade bladder epithelial cells. Various environmental cues, such as the availability of nutrients and the presence of host signals, modulate the expression of these virulence factors, allowing bacteria to adapt to the urinary tract environment and cause infection (Subashchandrabose et al. 2014; Hannan et al. 2012; Kostakioti et al. 2012). Moreover, a significant association between virulence factors of pathogens and antimicrobial resistance has been reported (Cepas and Soto 2020).

The significant risk to world health is increased by the development of antibiotic-resistant bacteria among uropathogenic strains. Therefore, knowledge regarding genome sequences of uropathogenic associated strains, presence of different virulence, drug resistance genes, identification of potential drug targets, and niche adaptations can aid to understand the pathogenicity of UTI and treat the patients properly. Thus, reducing the inappropriate use of antibiotics and utilizing shared proteins in subtractive proteomics techniques can suggest prospective therapeutic targets that may be further researched in the direction of personalized medicine.

Materials and methods

Sample collection

Midstream urine samples were collected from women included in the study. The samples were collected from Victoria Hospital Bahawalpur and Fatima Jinnah Medical University Lahore, Pakistan, based on criteria as described by Saraf et al. (2022)

with ethical approval. The samples were sent immediately to the microbiology laboratory, where they were cultivated and identified within 2 h to prevent any possible contamination or decrease in the number of microbes.

Inclusion criteria Premenopausal non-pregnant women (aged 41) with prior diagnosis of acute uncomplicated cystitis and pyelonephritis were approached. The diagnosis was also verified using the criteria given by the European Association of Urology Section of Infection in Urology classification of UTIs based on clinical presentation, risk factors, and severity scale. Regardless of their urban or rural background, samples were exclusively collected from women belonging to low socioeconomic group.

Exclusion criteria Women having pregnancy, menstruation, menopause, complicated UTIs, underlying diseases, prior antibiotic therapy, and middle and upper socioeconomic status were excluded from the study. The inclusion and exclusion criteria were followed as described by Smelov et al. (2016).

Isolation and identification of *E. coli*

Twenty samples were cultured on MacConkey agar (Oxoid, UK). Isolated *E. coli* colonies were obtained by selecting a single colony from the surface of the medium, re-streaking it on MacConkey agar plates, and then incubated the plates at 37°C for overnight. Biochemical tests, MR-VP, and Gram staining were used to investigate purified bacteria (Kurbasic et al. 2010), Lactose and Indole tests to confirm the presence of *E. coli*.

DNA extraction

The selected strain based on biochemical tests and following the manufacturer's instructions, genomic bacterial DNA (gDNA) was isolated using the DNeasy Blood & Tissue Kit (Qiagen). The Qubit dsDNA HS Assay Kit (Thermo Fisher Scientific, USA) was used to fluorometrically measure the DNA content. Agarose gel electrophoresis and spectrophotometry were used to confirm the material's integrity and purity respectively. To determine the genetic diversity and evolutionary relationship between *E. coli* strain CUI-B1 and those in other publicly available databases ($n = 1463$), we conducted phylogenetic analyses using 16S rRNA with the twenty most comparable strains based on BLAST search similarity (Feng et al. 2023). The tree derived from 16S rRNA has traditionally been used to define prokaryotic taxonomy. The genome sequence was conducted as described below.

Library preparation

Following the manufacturer's instructions, the Nextera XT Kit (Illumina) was used to prepare the library from the gDNA isolated from the verified sample for genome sequencing. Briefly, limited-cycle PCR was used to fragment and amplify 1 ng of DNA using Nextera XT barcodes (12 cycles). The Agencourt AMPure XP magnetic beads were used to purify the PCR product (Beckman Coulter). After diluting each library to 4.0 nM, they were pooled. To assess the quality of the libraries, DNA quantification was performed via Qubit and Agilent Bioanalyzer. Barcoded libraries that were pooled were diluted to 12 pM. Using a MiSeq Illumina (Macrogen South Korea), whole-genome sequencing was carried out.

Quality filtering, alignment, and de novo assembly

The Illumina adaptors, final sequences, and bases below Q20, and unnamed bases were removed from FastQ files during the purification process using cut adapt 1.4. The Q score and A, T, G, and C fractions were determined using the Rqc v1.10.2 R package. Bowtie v2.3.3.1 was used for the alignment of quality-filtered reads using *E. coli* O25b: H4-ST131 as reference, and Picard Tools v2.15.0 was used to sort the reads by chromosome coordinates. In case of overlapping reads, FLASH2 v2.2.00.21 was used to merge them into one read. De novo sequencing was performed with Genius Prime v3 to verify the detection of all genes absent from the previously identified *E. coli* strain.

Comparative genomic analysis

The genome comparative analysis of *E. coli* strain CUI-B1 with representative UTI-associated strains such as *E. coli*, *P. mirabilis*, *K. pneumoniae*, and *S. saprophyticus* strains was conducted using the BRIG software package (Alikhan et al. 2011). The tool is used to create alignment of several genomes to examine very similar subsequences, evolutionary processes like inversion, and rearrangement and comparison. Moreover, GC skew and GC content map were generated using CGView Server V1.0 (Grant and Stothard 2008).

Genome-wide gene prediction and analysis

To determine the virulence genes of the *E. coli*, *P. mirabilis*, *K. pneumoniae*, and *S. saprophyticus* genomes, we retrieved the FASTA gene sequences present in the Virulence Factor Database (VFDB). This VFDB version included more than 30178 genes, encoding over 1800 unique virulence factors (VFs), across over 30 distinct genera of bacteria. Nucleotide-Nucleotide BLAST v2.7.1+ was used to match the whole VFDB gene set with the genomes of the *E. coli* strain O25b:H4-ST131, *E.*

coli strain CUI-B1, *P. mirabilis* HI4320, *K. pneumoniae* strain U1, and *S. saprophyticus* strain ATCC 15305. We kept only the sequences that spanned 80% or more of the VFDB query gene. Only the VFDB gene that efficiently matched with the chosen genomes was taken into consideration when multiple VFDB genes displayed similar alignments to an assembly as described by Paniagua-Contreras et al. (2019). Based on VFDB, the 349 VFDB genes for which the orthologous genes were found were manually categorized into functional categories (Table S1).

Subtractive proteomic approach for prediction of potential drug targets

To predict the potential drug targets by subtractive proteomics approaches, the common virulence proteins which encode the key virulence features such as iron uptake, adherence, toxin, and regulating proteins were further processed extensively for the identification of potential drug targets. The flow chart of the methodology is shown in Fig. 1. Firstly, the elimination of homologous proteins from the shared virulence proteins was performed by BLASTp against human proteome as described by Ehsan et al. (2018). We have considered only those proteins with no hits found as non-homologous proteins after performing BLASTp against human proteome and kept for further analysis. To look into the location of the virulence proteins particularly to ascertain the suitability of drug targets, the subcellular localization of the final set of non-homologous proteins was predicted by using PSORTb (Yu et al. 2010). The proteins with cytoplasmic nature were further processed for the identification of unique metabolic pathways by KAAS server at KEGG. Unique metabolic pathways mean the pathways encoded by bacterial species but not humans. KAAS (KEGG Automatic Annotation Server) provides functional annotation of genes by BLAST comparisons against the manually curated KEGG GENES database. The result contains KO (KEGG Orthology) assignments and automatically generated KEGG pathways (Moriya et al. 2007). Proteins encoded by common metabolic pathways were discarded. In further screening, the drugability potential of proteins was assessed. Drugability is the probability of the proteins binding to a drug-like compound. For this purpose, each drug target protein was aligned with DrugBank and opted the proteins which have appropriate drug target either approved by FDA, experimental, nutraceuticals, or biotech drugs.

Results and discussion

Isolation and identification of *Escherichia coli*

Given that every year, urinary tract infections affect 150 million individuals throughout the globe (Flores-Mireles et al. 2015), all the strains employed in this study have clinical relevance for conditions like UTIs. Table 1 presents

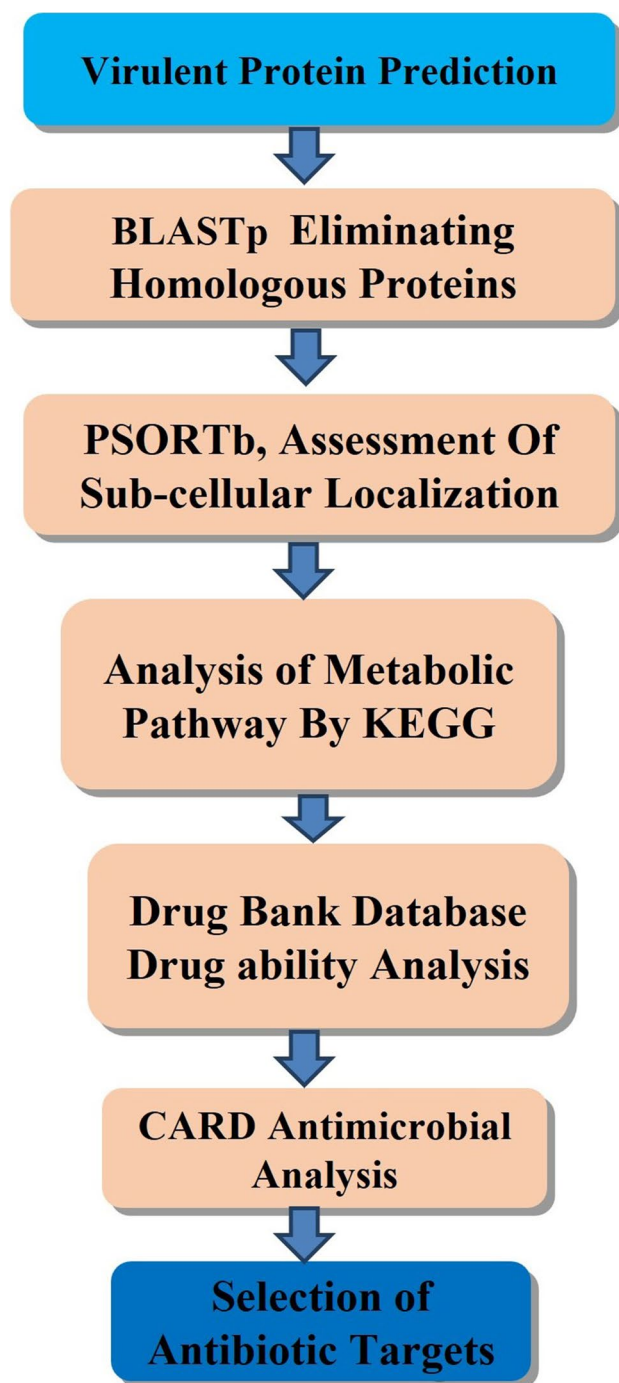


Fig. 1 Graphical presentations of the subtractive proteomics approaches for potential drug targets

20 potential *E. coli* isolates causing UTIs, 17 (85%) tested positive for indole production and methyl-red test however negative for urease production as well as Voges-Proskauer reaction. As described by Holt et al. (Holt et al. 1994), these isolates were confirmed to be *E. coli* and a randomly single isolate was further processed for 16s RNA sequencing, genome sequencing, and characterizations.

Evolutionary analysis of *E. coli* strain CUI B1

The 16S rRNA tree revealed that *E. coli* isolates CUI-B1 seemed genetically more identical to *E. coli* isolates such as *E. coli* strain SE, *E. coli* strain E1, and *E. coli* strain E10 than those from the public database (Fig. 2). *E. coli* strain E1 has been identified as the causative agent of various diseases in poultry, such as yolk sac infection, *E. coli* strain E10 caused multi-bacterial airsacculitis (Feng et al. 2023), and *E. coli* strain SE was reported in a public database as the causative agent of udder health and neonatal disease.

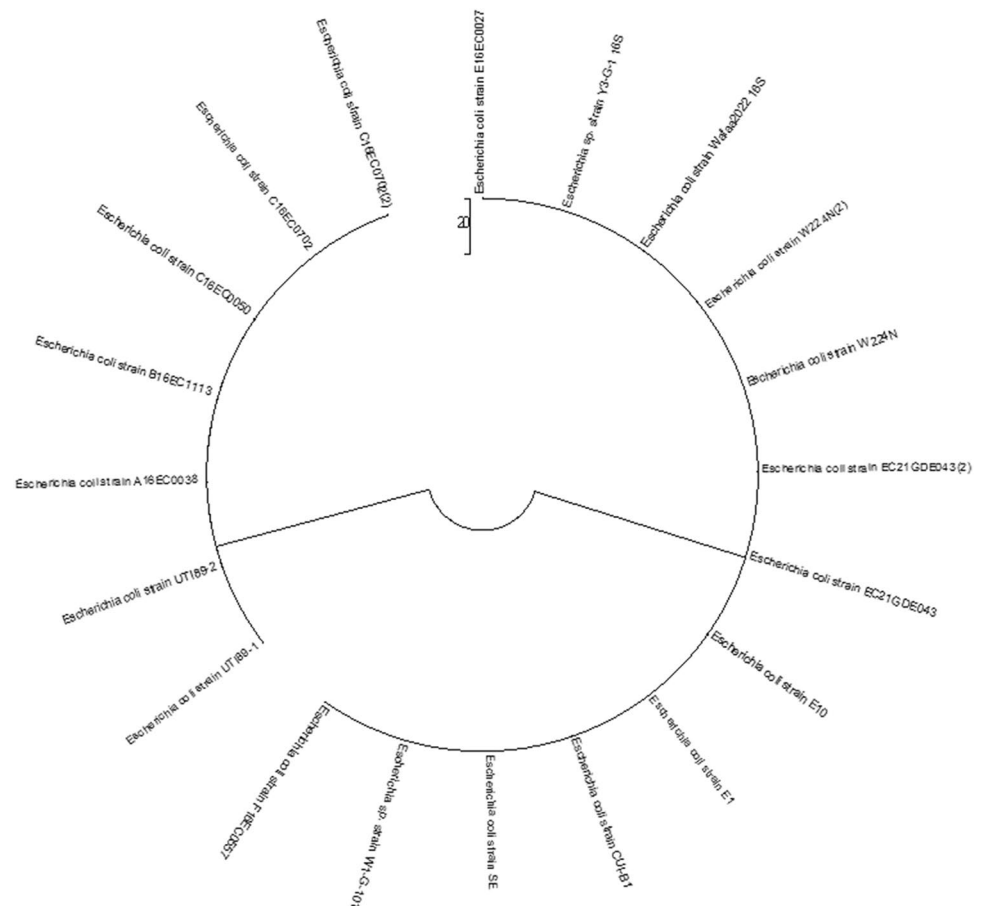
Several genotypes derived from *E. coli* strain UTI89-2 are shown in another cluster which is depicted in Fig. 2. Infections caused by *E. coli* transmitted through contaminated meat products are frequently associated with intestinal disorders, such as diarrhea. However, researchers have noted *E. coli*-related UTIs that seem to be connected to exposure to animals used in food production (Feng et al. 2023). The 16S rRNA gene is a highly conserved component of the transcriptional apparatus in all DNA-based life forms, and horizontal gene transfer has minimal effect on it (Daubin et al. 2003). Nonetheless, in literature, several variations in certain variable regions of the 16S rRNA gene have been reported, and in addition to these variations and phylogenetic analyses, the strain CUI-B1 was clustered in the same group as *E. coli* E1 and *E. coli* E10, while *E. coli* strain UTI89-2 belonged to a different cluster. Experimental studies demonstrated the shared pathogenic potential of human UTIs and avian pathogenic *E. coli*, suggesting that these extraintestinal *E. coli* may be descended from the same bacterial lineages or share common evolutionary origins (Kathayat et al. 2021). Consistent detection of specific human UTI lineages in poultry or poultry products, but rarely in other meat products, supports the notion that poultry may serve as a reservoir for human UTIs.

Genomic features of UTI-associated strains

Overall genomic features of these strains are as follows. The draft genome sequence of *E. coli* strain CUI-B1 which was isolated in our study is of 5.45-Mbp genome size and comprised of 12 contiguous DNA sequences with 6029 coding genes, 83 tRNA, 23 rRNA, and 50.5% GC contents. Similarly, the *E. coli* strain O25b: H4-ST131 which was used as a reference strain in this study encodes 5329 coding genes, 103 tRNA, 22 rRNA, 50.5% GC contents with single chromosome, and a total genomic length of 5.49 Mbp. The genome sequence of *P. mirabilis* strain H14320 causal agent of intestinal tracts of humans consists of single chromosomal DNA of 4.06 Mbp along with 1 plasmid. Its genome encodes 3752 genes, 83 tRNAs, and 22 rRNAs, with a GC content of 38.9% (Table 2). The genome sequence of *K. pneumoniae* strain 1721 possesses three contiguous DNA sequences and a plasmid collectively of

Table 1 Biochemical characterization of *E. coli*

Isolate No.	Indol test	Methyl red test	Voges. P test	Urease test	Acid from sugars	
					Glucose	Lactose
CUI-B1	+	+	—	—	+	+
CUI-B2	+	+	—	—	+	+
CUI-B3	+	+	—	—	+	+
CUI-B4	+	+	—	—	+	+
CUI-B5	+	+	—	—	+	+
CUI-B6	—	—	—	—	+	+
CUI-B7	+	+	—	—	—	—
CUI-B8	+	+	—	—	+	+
CUI-B9	+	+	—	—	+	+
CUI-B10	+	+	—	—	+	+
CUI-B11	+	+	—	—	+	+
CUI-B12	—	—	—	—	+	+
CUI-B13	+	+	—	—	+	+
CUI-B14	+	+	—	—	+	+
CUI-B15	+	+	—	—	+	+
CUI-B16	+	+	—	—	+	—
CUI-B17	+	+	—	—	+	+
CUI-B18	—	—	—	—	+	+
CUI-B19	+	+	—	—	+	+
CUI-B20	+	+	—	—	+	+

Fig. 2 16S rRNA gene sequence-based phylogenetic analysis of *E. coli*

5.46 Mbp. The genome encodes 5500 genes, 86 tRNAs, and 9 rRNAs, with a GC content of 57.29%. The draft genome sequence of *S. saprophyticus* strain ATCC 15305 consists of four contiguous DNA sequences and with three plasmids collectively of 2.59 Mbp, encoding 2522 genes, 60 tRNAs, and 13 rRNAs, with a GC content of 33.1% (Table 2).

Overall, the genomic features show a variation not only in the genome size, and GC contents but also in gene coding sequences. Most importantly, the amino acid composition which is shaped by GC contents is correlated with the protein structure classes, the metabolic efficiency, and the translation efficiency leading to the protein functioning varied in each species (Meng-Ze et al. 2018). These findings could be a potential source to understand the epidemiology, diversity, pathogenicity, and drug resistance characteristics following potential anti-virulent drug targets (Table 3).

Comparative genome analysis

For comprehensive genome-wide comparative analysis, the results of the sequence-based comparison were obtained for *E. coli*, *K. pneumoniae*, *P. mirabilis*, and *S. saprophyticus* strains used in this study by using the RAST server and BRIG genome-wide analysis tools. The *E. coli* strain O25b: H4-ST131 was used as a reference for visualization of the genomic regions, which are shown in Fig. 1. The circular genomes of the *E. coli* strain CUI-B1,

K. pneumoniae strain 1721, *P. mirabilis* strain H14320, and *S. saprophyticus* strain ATCC 15305 unveiled the % age of GC contents, coding sequence (CDS), open reading frame, and total no. of RNAs (Fig. 3). The visualization analysis of the entire genomes presented significant similarity levels (>90%) between *E. coli* strain CUI-B1 and *E. coli* strain O25b: H4-ST13, while similarly with the rest of the genomes was ranging from 40 to 70% and numerous positions of latent insertion/deletion sites were depicted in the genomes of *E. coli*, *K. pneumoniae*, *P. mirabilis*, and *S. saprophyticus* strains. Moreover, various hypothetical genes were encoded by these strains consistent with previous genome-wide studies. In genomics, the comparisons and characterizations of functional elements depend on the identification of the natural selection footprints (Kim et al., 2012). Such studies reveal a new dimension of comparative genomics such that a comparison of five genomes associated with UTI highlights comprehensively the differences in overall levels of genome organization.

Identification of antimicrobial resistance genes

In the five genomic assemblies, the resistant gene identifier predicted the wide distribution of various antibiotic-resistant genes among all strains. In the analysis of five different genomic assemblies, the resistant gene identifier was used to predict the presence of various antibiotic-resistant

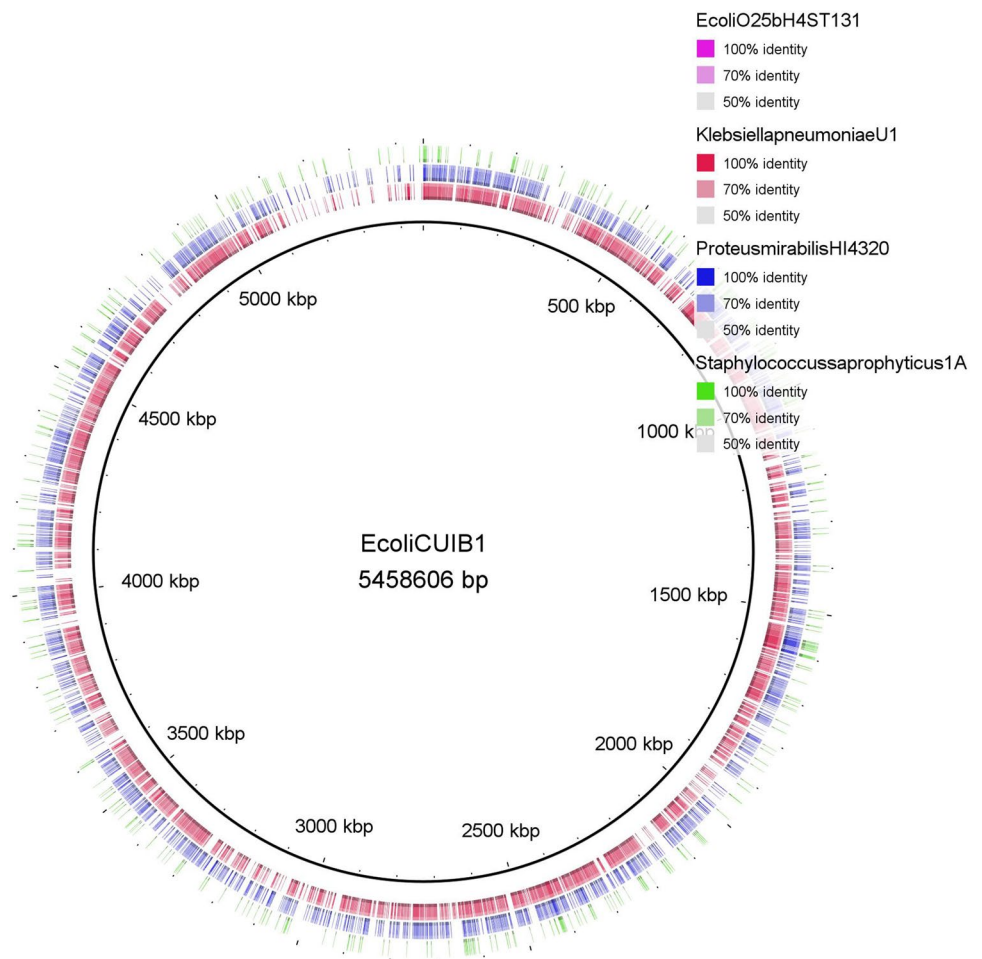
Table 2 Genomic features of strains associated with urinary tract infection

	<i>Escherichia coli</i> strain CUI-B1	<i>Escherichia coli</i> O25b:H4-ST131	<i>Proteus mirabilis</i> strain H14320	<i>Klebsiella pneumoniae</i> strain 1721	<i>Staphylococcus saprophyticus</i> strain ATCC 15305
Origin	Human	Human	Human	Human	Human
Contigs No.	12	1	1	3	1
Genome size (Mb)	5.38	5.40	4.06	5.46	2.59
Coding genes	6029	5329	3752	5500	2522
G+C content (%)	50.4	50.5	38.9	57.29	33.1
tRNA No.	83	103	83	86	60
rRNA No.	23	22	22	9	13

Table 3 Drugability potential of the five druggable targets

Name of gene	Drug Bank ID	Drug Bank target	Drug group	Molecular weight (kDa)	Drug resistance
entB	DB01942/DB02793	Putative hydroxy pyruvate isomerase YgbM/Henazine biosynthesis protein PhzD	Experimental, Investigation	32.0254	Ethambutol
clbH	DB01672	2,3-Dihydroxy-benzoic acid	Experimental	83.12	Chloramphenicol
chuV	DB00171	ATP	Investigational, Nutraceutical	45.62	Colistin
ybtS	DB01942	Putative hydroxypyruvate isomerase YgbM	Experimental, Investigational	32.02	Metallo- β -lactamase

Fig. 3 Comparative genome visualization of uropathogenic bacteria



genes across all strains. For instance, the *E. coli* strain CUI-B1 was found to have 56 antibiotic resistance genes. These genes confer resistance to several types of antibiotics, such as β -lactams, fluoroquinolones, aminoglycosides, sulfonamides, chloramphenicol, trimethoprim, and tetracycline. This is in comparison to the reference *E. coli* strain O25b:H4-ST131, which has 57 different types of antibiotic-resistant genes. Similarly, the *K. pneumoniae* strain 1721 carries a certain number of such genes, while the *P. mirabilis* strain HI4320 carries 14, and the *S. saprophyticus* strain ATCC 15305 carries 10 antibiotic-resistant genes (Table S1). Overall, each strain encodes at least six antibiotic-resistant genes. The genes with the highest prevalence among the strains were *rsmA*, *KpnF*, and *CRP*, the only three genes found in all strains from different genera. *rsmA* is a gene that regulates the virulence of *P. aeruginosa*. However, its negative effect on MexEF-OprN overexpression has been noted to confer resistance to various antibiotics (Pessi et al. 2001) and its homolog in *E. coli* is *csrA*. The *kpnF* gene found in all strains except *S. saprophyticus* strain ATCC 15305 which

is previously described in *K. pneumoniae* also boosts high resistance to different antibiotics, antiseptics, and disinfectants (Srinivasan and Rajamohan 2013). The acute-phase protein, also called C-reactive protein or CRP, is an acute-phase protein and the name of the protein is due to its ability to precipitate the *S. pneumoniae* C-polysaccharide. CRP is potentially a useful tool to enhance antibiotic stewardship (Cals and Ebell 2018). *rsmA*, *KpnF*, and *CRP* exhibited a stronger genetic relationship between strains that had profiles of gene-producing heavy metal efflux pumps regardless of their origin.

The high incidence of antibiotic genes that code for drug resistance serves as an alert to us regarding the necessity of rigorous surveillance programs to track resistance and the spread of these diseases among various sources. Moreover, antibiotic resistance gene profiling of strains studied here provides complexity of antimicrobial resistance (AMR) transmission and movement proclivity. On the other hand, compared to other strains, *E. coli* had much higher odds of being identified when resistance determinants were

examined by species type as they offer some more hints regarding possible initial sources and the trafficking of specific resistance genes. In short, one of the biggest obstacles to effectively treating UTI infections in the clinic is the composition and distribution of antibiotic resistance genes.

Detection of virulence genes and analysis

Using the complete dataset of virulent genes retrieved from the VFDB by using various keywords (Table S2), BLAST analysis was performed against each genome assembly and found 85 to 205 virulence genes in the five genome assemblies. Such as *E. coli* strain O25b: H4-ST131 encodes 208 virulent genes, *E. coli* strain CUI-B1 encodes 205, *P. mirabilis* strain H14320 comprised 202, *K. pneumoniae* strain 1721 comprised 152, and *S. saprophyticus* strain ATCC15305 comprised 85 virulent genes distributed at various genomics regions, while 44 of them were shared by all the UTI-associated strains (Table S3). The most abundant among these shared genes included the genes which are linked in iron uptake such as Ferric enterobactin transport protein FepE, iron/manganese transport gene SitC, enterobactin exporter, and salicylate synthase Irp9. For almost all living cells, iron is a vital micronutrient. Both invasive pathogens and human cells, including those of the immune system, need iron to maintain their function, proliferation, and metabolism. On the one hand, most human diseases' pathogenicity is associated with microbial iron uptake (Nairz and Weiss 2020). Contrarily, the sequestering of iron from bacteria and other pathogens is a successful host defense tactic that adheres to the fundamentals of nutritional immunity (Nairz and Weiss 2020).

Because free iron is bound to heme or circulated proteins like transferrin or lactoferrin, bacterial pathogens cannot access it. Pathogens obtain iron from the host by a variety of processes, including the uptake of Fe²⁺ (Feo system), the absorption of heme, and the production of extracellular Fe³⁺-chelating molecules known as siderophores, which are either produced by the pathogens themselves or by other bacteria (Caza and Kronstad 2013; Cornelis and Dingemans 2013). So, the presence of virulent genes associated with iron uptake in a pathogen can indeed be a cause for concern as it indicates a potential mechanism for the *E. coli* strain CUI-B1 to obtain nutrients from the host and potentially cause disease.

Following iron uptake-associated genes, various types of toxin genes were encoded by all selected strains of UTI. Among them were ket toxin genes, uropathogenic-specific protein (usp), acyl-CoA dehydrogenase (clbF), malonyl-CoA transacylase (clbG), thioesterase (clbQ), putative amidase (clbL) etc. Gram-negative enteric pathogens, such as *E. coli*, are particularly adept at acquiring virulence factors through HGT, which has contributed to their ability to cause a wide range of infections in humans (Ochman et al. 2000). The uropathogenic-specific protein (Usp) of *E. coli* is

a genotoxin that is toxic to mammalian cells and is linked to strains that cause prostatitis, pyelonephritis, and bacteremia. The prevalence of usp in *E. coli* strains from patients with colorectal cancer has also been reported (Rihtar et al. 2020). The identification of usp in this study in both gram-negative and positive bacteria seems to be the linked with HGT.

Another group of proteins distributed among the UTI genomes was proteinaceous toxins, such as putative thioesterase, putative malonyl-CoA transacylase, and putative acyl-CoA dehydrogenase. Adhesins are found on the surface of bacterial cells that aid in the process of adhering to other cells or inanimate surfaces (Shruthi et al. 2012). To successfully colonize a new host, bacteria must first attach to its surface. The VFs on pathogenic bacteria can be used as indicators and antibacterial targets. However, various UPEC strains might take shelter in multiple VFs, and numerous factors are only found in a portion of UPEC strains (Shah et al. 2019). Hence, it appears that the development of efficient and generally applicable strategies to prevent or cure UTIs brought on by UPEC requires a combination of several VFs found in the currently under study strains. The creation of such cutting-edge approaches to managing illness would be made easier by identifying the pattern in which VFs cause UTIs and comprehending their epidemiological spread.

In short, the virulence factors identified in our research included features from both databases and had at least 90% sequence similarity with reference genes. UTI virulence genes include temperature-sensitive haemagglutinin (tsh), ferric aerobactin receptor (iutA), increased serum survival (iss), heat-resistant agglutinin (hra), P fimbrial adhesin (papC), capsular polysialic acid virulence factor group 2 (kpsII), colicin V (cvaC), and invasive factor of brain endothelial cells locus A (ibeA) of *E. coli* strains. The advent of drug-resistant microorganisms among UPEC strains with the distribution of virulent factors among entire genomes increases the global health risk (Tabasi et al., 2015). Overall, the distribution of virulence genes among the *E. coli* genome is complex and can vary depending on the strain and the specific virulence factors involved. Understanding the distribution and function of these virulent genes is an active area of research and is important for developing effective strategies for the prevention and treatment of *E. coli* infections.

Subtractive proteomics approach for prediction of potential drug targets

By using various computational analyses, 44 common virulent proteins were further analyzed and considered as suitable targets for the formulation of novel anti-virulent drugs. Anti-virulence intends to selectively demobilize pathogens that slow infection rates. Anti-virulence drugs will maintain the endogenous microbiome while exposing pathogens susceptible to elimination through the host immune system, in contrast to

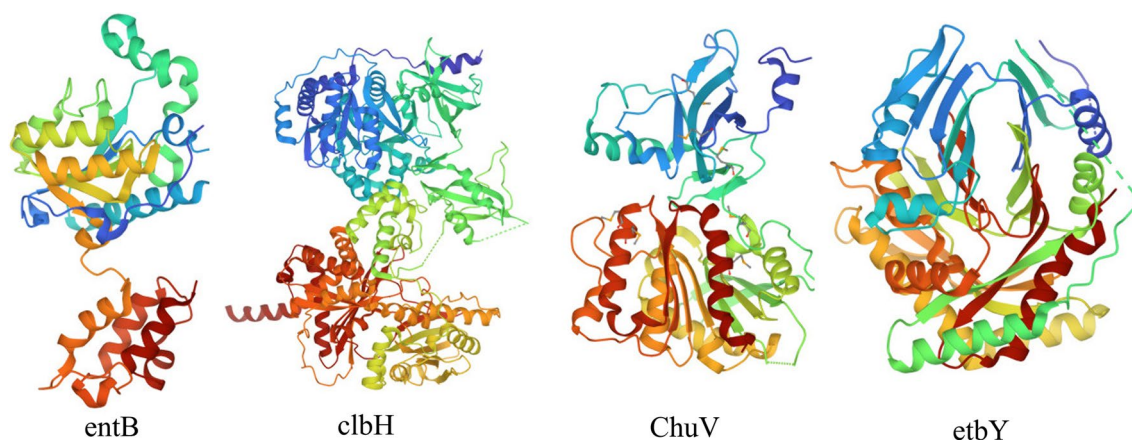


Fig. 4 The 3D structure of potential drug targets

the fact that conventional antibiotics do not generally differentiate between commensal intestinal bacteria and pathogenic microbes (Knowles and Gromo 2003). We ran BLASTp on the human proteome and identified 38 proteins as non-homologous out of 44 shared proteins to exclude host homolog from the non-redundant proteome. Host non-homologous proteins are proteins that are not similar in structure or sequence to proteins produced by the pathogen or other foreign antigen. Because these proteins are not like foreign antigens, they are less likely to trigger an immune response against the host's tissues (Naz et al. 2015; Sanober et al. 2017). Thirty-eight non-homologous proteins (Table S4) were analyzed for localizations and results revealed that 22 proteins were the proteins belonging to cytoplasmic locations (Table S5). Predicting the cellular localization of unknown proteins also allows researchers to investigate their activity, participation in disease mechanisms, and potential therapeutic development pathways. Drugs have an easier time accessing cytoplasmic proteins than membrane proteins; hence, they are favored (Reygaert 2018; Sanober et al. 2017).

Virulent, cytoplasmic proteins were examined in KAAS for their corresponding metabolic pathways (Moriya et al. 2007) which may successfully identify the collection of pathogen-specific proteins required for the pathogen's survival (Solanki and Tiwari 2018; Uddin et al. 2015). Out of the 21 protein sets, 14 proteins are members of unique metabolic pathways, and the bulk (80%) of these proteins are members of several pathogen-distinct metabolic pathways. In Table S6, specifics of distinctive metabolic pathways are presented. The likelihood of side effects is lower for proteins that belong to distinct pathways since there are no competing pathways in the host, making them the most likely drug targets.

The ability of a protein to bind small drug-like molecules with a high affinity is referred to as druggability. For this purpose, a Drug Bank database was used in which BLASTp was performed on each drug target with a set of criteria against a database of substances from DrugBank: Experimental drugs, FDA-approved, nutraceuticals, small molecules, and biotech.

Based on a bit score > 100, 5 of the initial 14 proteins were selected as possible pharmaceutical targets and belonged to the drug group experimental small-molecule medications, exploratory or FDA-approved. In further characterizations, the molecular weight of all potential pharmacological targets was predicted by the ExPASy ProParam tool (Gasteiger et al. 2005). Proteins with a molecular weight of 110 kDa or less are suggested for use in the current study because they are easy to purify (Naz et al. 2015). Proteins selected for the drug have a molecular weight of 32 to 84 kDa, providing evidence of the value of protein screening for drug targets.

Resistance was also considered as a standard for choosing which druggable proteins to be prioritized as described previously (Adamus-Bialek et al. 2013; Ehsan et al. 2018). Literature revealed that *Escherichia coli* O25b:H4-ST131, *P. mirabilis* strain HI4320, *K. pneumoniae* strain 1721, and *S. saprophyticus* strain ATCC 15305 exhibited resistance to multiple antibiotics such as fosfomycin, aminoglycosides, fluoroquinolones, tetracycline, and sulfonamides. Putative amidase clbL was the only protein that did not exhibit drug resistance and was discarded. Moreover, all the proteins have a suitable template for the prediction of the structure of each protein (Fig. 4), therefore, seems to be considered a potential drug target for *E. coli* strain CUI-B1, *E. coli* strain O25b: H4-ST131, *P. mirabilis* strain HI4320, *K. pneumoniae*, and *S. saprophyticus* strain ATCC15305.

Conclusions

The evolutionary analysis demonstrated the shared evolutionary cluster of human urinary tract infections and avian pathogenic *E. coli*, suggesting that these extraintestinal *E. coli* may be descended from the same bacterial lineages or share common evolutionary origins. Consistent detection of specific human UTI lineages in poultry or poultry products, but rarely in other meat products, supports the notion that

poultry may serve as a reservoir for human UTI. Moreover, the advent of drug-resistant bacteria in UTI-associated strains increases the serious threat to global health. Genome sequence and comparative analysis revealed *E. coli* strain CUI-B1 seems to be among the group of serious threats to healthcare settings that need to be addressed well in time. Moreover, based on genome sequence, virulence, and antimicrobial-resistant gene dataset following reverse engineering approach revealed 22 proteins mapped to pathogens unique pathways and characterized four proteins entB, clbH, chuV, and ybtS to be a potential druggable target which could be utilized for these strains. Identifying potential drug targets is an important aspect of drug discovery and development. The findings may provide the foundation for suitable targets for prospective drug discovery for *E. coli*, *P. mirabilis*, *K. pneumoniae*, and *S. saprophyticus*-based infections and could be employed in personalized drug development also.

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Author contributions AA, RN, and AI collected samples, performed basics analysis, and wrote 1st relevant 1st draft. AH, WBA, AH, and IF performed and guided bioinformatics analysis. ZMZ wrote the “Discussion” section and proof reading. MI supervised whole work and drafted the manuscript for final submission.

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Data availability The 16s RNA sequence data was submitted to GenBank/DDBJ/ENA under accession no. OQ858381, while the Whole Genome Shotgun project has been deposited at DDBJ/ENA/GenBank under the accession JARWMK000000000. The version described in this paper is version JARWMK010000000.

Declarations

Conflict of interest The authors declare no competing interests.

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